# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 9 — MATHS FULL SYLLABUS PAPER SET 09

**Full Syllabus (Chapters 1–8)** | RBSE Pattern | Session 2026–27

**Time: 3 Hours 15 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 80**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 20 × 1 = 20

*Choose the correct option and write it in your answer book.*

**1.** The midpoint of the segment joining $A(2,\ 4)$ and $B(6,\ 10)$ is: 1

**(a)** $(8,\ 14)$ **(b)** $(4,\ 7)$ **(c)** $(4,\ -7)$ **(d)** $(2,\ 3)$

**2.** The distance of the point $(6,\ -8)$ from the origin is: 1

**(a)** 6 units **(b)** 10 units **(c)** 11 units **(d)** 14 units

**3.** The point $(-3,\ 5)$ lies: 1

**(a)** Quadrant III **(b)** Quadrant IV **(c)** Quadrant I **(d)** Quadrant II

**4.** The coefficient of $x^2$ in $x^4 - 3x^3 + 6x^2 - 2x + 7$ is: 1

**(a)** −2 **(b)** −3 **(c)** 7 **(d)** 6

**5.** The sequence $1, 7, 13, 19, \ldots$ forms a linear pattern. Its $n$th term is: 1

**(a)** $6n - 5$ **(b)** $6n + 1$ **(c)** $6n - 7$ **(d)** $n + 6$

**6.** The number $0.\overline{3}$ in the form $\frac{p}{q}$ is: 1

**(a)** $\frac{1}{3}$ **(b)** $\frac{13}{30}$ **(c)** $\frac{2}{3}$ **(d)** $3$

**7.** The decimal expansion of $\frac{11}{40}$ is: 1

**(a)** non-terminating non-repeating **(b)** terminating

**(c)** non-terminating repeating **(d)** cannot be determined

**8.** Brahmagupta defined zero as the result of: 1

**(a)** multiplying by 1

**(b)** subtracting a number from itself ($a - a = 0$)

**(c)** adding two equal numbers

**(d)** dividing a number by itself

**9.** The expression $49x^2 + 28xy + 4y^2$ can be factorised as: 1

**(a)** $(7x + 2y)^2$ **(b)** $(49x + 2y)^2$ **(c)** $(7x - 2y)^2$ **(d)** $(7x + 4y)^2$

**10.** A student writes $(a + b)^2 = a^2 + b^2$. The correct response is: 1

**(a)** Correct only for $a > b$ **(b)** Correct for all positive $a, b$

**(c)** Correct for all $a, b$ **(d)** Incorrect; the term $2ab$ is missing

**11.** ABCD is a cyclic quadrilateral with $\angle A = 110^\circ$. Then $\angle C$ is: 1

**(a)** $35^\circ$ **(b)** $250^\circ$ **(c)** $110^\circ$ **(d)** $70^\circ$

**12.** In a circle of radius $13$ cm, a chord is $5$ cm from the centre. The length of the chord is: 1

**(a)** 22 cm **(b)** 24 cm **(c)** 12 cm **(d)** 26 cm

**13.** An arc subtends an angle of $84^\circ$ at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

**(a)** $138^\circ$ **(b)** $84^\circ$ **(c)** $168^\circ$ **(d)** $42^\circ$

**14.** The area of a parallelogram with base $12$ cm and height $7$ cm is: 1

**(a)** 84 cm$^2$ **(b)** 38 cm$^2$ **(c)** 96 cm$^2$ **(d)** 42 cm$^2$

**15.** Taking $\pi = \frac{22}{7}$, the area of a circle of radius $3.5$ cm is: 1

**(a)** 19.25 cm$^2$ **(b)** 38.5 cm$^2$ **(c)** 77 cm$^2$ **(d)** 22 cm$^2$

**16.** The length of an arc subtending $60^\circ$ at the centre of a circle of radius $21$ cm ($\pi = \frac{22}{7}$) is: 1

**(a)** 132 cm **(b)** 11 cm **(c)** 44 cm **(d)** 22 cm

**17.** A bag contains 7 red, 5 blue and 8 green balls. One ball is drawn at random. The probability that it is green is: 1

**(a)** $\frac{1}{5}$ **(b)** $\frac{3}{5}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{1}{3}$

**18.** A sample shows a probability of $\frac{3}{10}$ that a student likes cricket. In a school of 2000 students the expected number who like cricket is: 1

**(a)** 700 **(b)** 800 **(c)** 300 **(d)** 600

**19.** The 10th triangular number is: 1

**(a)** 50 **(b)** 55 **(c)** 45 **(d)** 66

**20.** For the sequence $t_n = n^2 - 2n + 3$, the value of $t_7$ is: 1

**(a)** 37 **(b)** 76 **(c)** 39 **(d)** 38

## SECTION — B (Very Short Answer Type) 8 × 2 = 16

*Answer in about 20–30 words each.*

**21.** Find the distance between $A(-3,\ -3)$ and $B(5,\ 3)$. 2

**22.** A linear pattern begins $7,\ 11,\ 15, \ldots$ Write the expression for its $n$th term and find the 15th term. 2

**23.** Without long division, state whether $\frac{11}{20}$ has a terminating or a repeating decimal. Give a reason. 2

**24.** Factorise $49x^2 + 14xy + y^2$. 2

**25.** A chord of length 40 cm is 21 cm from the centre of a circle. Find the radius. 2

**26.** Find the area of a triangle with sides 7 cm, 15 cm and 20 cm. 2

**27.** A letter is picked at random from ‘PROBABILITY’. Find the probability of 2

(i) the letter B

(ii) a vowel.

**28.** Find the first five terms of the sequence $t_n = 3n - 4$. 2

## SECTION — C (Short Answer Type) — 8 × 3 = 24

*Answer in about 40–60 words each.*

**29.** Find the length of the hypotenuse and the area of the triangle with vertices $O(0,0)$, $A(6,0)$ and $B(0,8)$. 3

**30.** The points $(-3,6)$, $(-2,4)$, $(0,0)$, $(1,-2)$ lie on a line. Find its equation and its slope. What does the slope tell? 3

**31.** Find the rational number $x$ such that $\frac{x}{3}+\frac{x}{2}=\frac{16}{3}$. 3

**32.** Use $(x-y)(x^2+xy+y^2)=x^3-y^3$ to factorise $27a^3-8b^3$. 3

**33.** A wheel has diameter 56 cm. How many revolutions will it make to cover 1.76 km? ($\pi=\frac{22}{7}$) 3

**34.** A class of 40 has 24 girls. One student is chosen at random. Find the probability that the student is 3
(i) a girl
(ii) a boy.

**35.** A triangle ABC is inscribed in a circle with centre O and $\angle BAC=50^\circ$. Find $\angle BOC$ and $\angle OBC$. 3

**OR**

ABCD is a cyclic trapezium with $AB \parallel CD$. Show that its non-parallel sides AD and BC are equal. 3

**36.** A child arranges marbles in rows: 1 in the first row, 2 in the second, … up to 25 rows. How many marbles are used? 3

**OR**

Find $12+13+14+\cdots+40$ using $S_n=\frac{n(n+1)}{2}$. 3

## SECTION — D (Long Answer Type) — 4 × 5 = 20

*Answer in about 100–150 words each.*

**37.** (i) Show that between any two rational numbers there is another rational number. 5
(ii) Find the rational number midway between $-\frac{1}{4}$ and $\frac{5}{12}$ and place it on the number line (describe).

**OR**

**Case study:** Mādhava's series gives $\pi=4\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots\right)$ and Āryabhaṭa gave $\pi\approx 3.1416$. 5
(i) Is $\pi$ rational?
(ii) Compute $4\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}\right)$ and compare with 3.1416.
(iii) Name the mathematician who proved π irrational and the year.

**38.** Prove that the angle subtended by an arc at a point on the circle is half the angle at the centre in the case where the line DC produced meets the circle at a point E outside the arc. 5

**OR**

**Case study:** A ground is a circle of radius 13 m. Two parallel boundary ropes (chords) are 10 m and 24 m long. 5
(i) Find the distance of each from the centre.
(ii) How far apart are they if on opposite sides of the centre?
(iii) How far if on the same side?
(iv) Which rope is longer and which is nearer to the centre?

**39.** **Case study:** A cyclic plot has sides 25 m, 39 m, 52 m and 60 m. 5

(i) Find its semi-perimeter.

(ii) Find its area by Brahmagupta's formula.

(iii) Show that the plot's diagonal of length 65 m is a diameter by finding the area as two right triangles (25-60-65 and 39-52-65).

(iv) Find the cost of levelling at ₹20 per $m^2$.

**OR**

**Case study:** A pizza of radius 14 cm is cut into 8 equal sectors. ($\pi = \frac{22}{7}$) 5

(i) Find the angle of each sector.

(ii) Find the arc length of one slice.

(iii) Find the perimeter of one slice.

(iv) Find the area of one slice.

**40.** **Case study:** A colony of 500 bacteria doubles every hour. 5

(i) Write the numbers at the end of hours 1 to 4.

(ii) Write the nth term.

(iii) How many bacteria after 6 hours?

(iv) After how many hours does the number first exceed 10,000?

**OR**

(i) Show that the sum of the first $n$ odd numbers is $n^2$. 5

(ii) Find $1 + 3 + 5 + \cdots + 49$.

(iii) Express 144 as the sum of consecutive odd numbers starting at 1.

**END OF QUESTION PAPER** **Total: 80 Marks**

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